

Semantic Compasses: Rank-2 Causal Dials in Attention-Head OV Singular Planes

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Abstract

Singular directions of an attention head’s W_{OV} map carry interpretable signal individually (Elhage et al., 2021; Millidge and Black, 2022; Ahmad et al., 2025); we ask whether *pairs* of these directions form a coherent two-dimensional causal structure. We define a *semantic compass* as a plane $\text{span}(u_i, u_j)$ on which (a) vocabulary projection yields antipodal categorical poles, (b) the residual injection $\alpha\sigma(\cos\theta u_i + \sin\theta u_j)$ at the head’s own layer produces a sinusoidal logit response with stable phase across α , and (c) the rotation cycle closes (2π -cyclic decode). On Phi-3 Mini, the same recipe yields a four-way temporal dial (month / hour, with second-harmonic ratio $A_2/A_1 = 0.32$ at $\alpha = 10$) and an entity-type dial. Across four pretrained decoders (GPT-2 Small, Phi-3 Mini, Gemma-2-2B, Llama-3.2-3B) we find gender compasses with phase drift $\leq 3^\circ$.

Phase stability and 2π -cyclic decode are the substantive evidence; amplitude linearity follows partly from the linear injection. Controls calibrate scope: the permutation null rejects label-coincidence ($p < 0.001$), while the random-plane null locates gender in a distributed 2D subspace rather than on a unique axis ($p > 0.10$ on every head, including under family-wise correction). On held-out CrowS-Pairs, dev-selected multipliers land Phi-3 gender at test $|\text{SS} - 50| = 5.73$ ($M^* = 0.50$) and select $M^* = 0$ on near-neutral baselines (GPT-2 race, Gemma gender), giving an off-switch that fixed projector baselines lack.

1 Introduction

A line of mechanistic-interpretability work establishes that individual singular directions of attention-head weights carry interpretable signal. Elhage et al. (2021) formalise the OV decomposition $W_{OV} = W_V W_O$ that factors a head into its read- and write-side maps; Millidge and Black (2022) show that singular directions of W_{OV} and

W_{QK} frequently decode through W_U to coherent vocabulary; Ahmad et al. (2025) extract sparse direction-level circuits via learnable masks over the singular values for the Gender Pronoun (GP), IOI, and Greater-Than tasks.

Several phenomena that language models exhibit do not reduce to single-axis binary contrasts. Temporal units (*month, hour, year*) form a four-way categorical structure; gender-marking heads can split singular vs. plural and masculine vs. feminine in non-orthogonal ways (Engels et al., 2024; Park et al., 2024); the same categorical axis can be *written* at one head and *steered* at another. We propose that reading attention-head weights as multi-dimensional geometric objects — starting with 2D planes spanned by pairs of singular directions — captures these phenomena directly. On such a plane an angle parameterises a continuous family of causal interventions, a radius parameterises magnitude, and the second harmonic separates two-way antipodal axes from four-way dials.

The compass. Given $W_{OV} = U\Sigma V^\top$ and writing $u_i := V_{[i,:]}^\top$, a *semantic compass* is an ordered pair (u_i, u_j) that passes three tests: (i) vocabulary decoding through W_U assigns antipodal categorical content at the two poles; (ii) the residual injection $v(\theta, \alpha) = \alpha\sigma(\cos\theta u_i + \sin\theta u_j)$, applied at the head’s own layer on the last token, produces a sinusoidal logit response whose phase is stable across α ; (iii) cycling θ from 0 to 2π closes onto identical decode vocabulary, with the antipode disjoint. The recipe is probe-free and weight-localized.

Why the geometry matters. Reading singular directions individually is a one-dimensional view of a higher-dimensional object. Two phenomena in our data are invisible to that view and surface only once we treat the plane as the unit of analysis.

Four-way categorical structure. On the Phi-3 L24H17 plane, a four-token probe at $\theta \in$

$\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ produces a strong second harmonic ($A_2/A_1 = 0.32$ at $\alpha = 10$); the same probe on the Phi-3 L24H28 plane gives $A_2/A_1 \leq 0.06$. The first encodes a four-way temporal dial (month / hour / year / upon); the second carries two superposed two-way axes. Reading either plane’s two singular directions one at a time gives the same vocabulary in both cases; the harmonic geometry distinguishes them.

Storage versus steering as a quantitative dissociation. The α -sweep slope and the in-plane causal-patch magnitude dissociate across heads (Table 4). Phi-3 L24H10 has the largest in-plane signal but a small sweep slope; L28H1 has the reverse profile. The same gender axis sits at two distinct operating points in the same model.

A near-causal plane evidenced across architectures. Five independent tests on every primary head support the plane as a near-causal axis: a sinusoidal α -sweep with phase drift $\leq 3^\circ$, 2π -cyclic decode that closes onto antipodal vocabulary at $\theta = 0$ and $\theta = 2\pi$, a permutation null at $p < 0.001$, an in-plane causal-patch test that flips logits in the predicted direction, and a blind causal scan that rediscovered the same heads without consulting decode. The same five tests pass on GPT-2 Small, Phi-3 Mini, Gemma-2-2B, and Llama-3.2-3B — two orders of magnitude in parameters and three distinct training pipelines. We stop short of “causal axis” because the random-plane null returns $p > 0.10$, with random-plane mean signals clustering at 20–30. We read this as gender being encoded over a distributed 2D subspace of the late-layer residual; the OV-SVD compass is a weight-readable coordinate system inside that subspace rather than its unique axis. Amplitude linearity in α ($R^2 \geq 0.99$) is consistent with the linear injection itself; phase stability is the non-trivial test that distinguishes a causal plane from a generic 2D direction.

Downstream behaviour. On held-out CrowS-Pairs (50/50 split, dev-selected multiplier M^* scored on test), Phi-3 gender at $M^* = 0.50$ lands at test $|\text{SS} - 50| = 5.73$, against a dev-half baseline of $|\text{SS} - 50| = 8.78$ at $M = 0$. The procedure also admits $M^* = 0$, dev-selected on 10 of 16 cells; on cells whose dev baseline is genuinely near 50 this generalises to small test $|\text{SS} - 50|$ (GPT-2 race 0.00, Gemma gender 0.38, Llama race 2.71, Llama gender 3.44). Fixed-projector debiasing methods have no analogous null option.

Contributions.

1. **The compass primitive.** A probe-free 2D causal dial read from W_{OV} by SVD, with a three-test calibration recipe (decode antipodes, phase-stable α -sweep, 2π -cyclic decode) and a five-test pre-registered falsification battery (§3).
2. **Cross-architecture replication on four model families.** Compasses with phase drift $\leq 3^\circ$ across $\alpha \in \{1, 3, 10\}$, permutation null $p < 0.001$, and 2π -cyclic antipodal decode hold on GPT-2 Small (124M), Phi-3 Mini (3.8B), Gemma-2-2B (2.6B), and Llama-3.2-3B (3.2B) — two orders of magnitude in parameters and three distinct training recipes (§4.2, Tables 1 and 3).
3. **Geometric structure beyond rank-1.** A four-way temporal dial in Phi-3 L24H17 with second-harmonic $A_2/A_1 = 0.32$ at $\alpha = 10$ vs. ≤ 0.06 on the rank-2 entity dial (§4.1); a quantitative storage-vs-steering dissociation across heads (§4.4); and two distinct compasses on the same model’s gender axis at near-orthogonal principal angles ($86.4^\circ, 89.0^\circ$ between Phi-3 L24H10 and L28H1, §4.5).
4. **A held-out routed downstream procedure.** Per-domain head routing, SNR-calibrated α , and dev-selected multiplier with $M = 0$ as a valid choice; the calibration recipe makes per-cell α comparable across models that differ by two orders of magnitude in singular-value scale (§5.3, Table 8).

2 Background and related work

OV circuit and singular-direction analysis. For a decoder head, $W_{OV} = W_V W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ (Elhage et al., 2021). With residuals as row vectors the head adds rW_{OV} to the destination token; taking $W_{OV} = U\Sigma V^\top$ gives $rW_{OV} = rU\Sigma V^\top$, so the rows of $V^\top$ are the write-out directions in residual coordinates. We let $u_i := V_{[i,:]}^\top$. Millidge and Black (2022) document that these singular directions are often individually interpretable through W_U . Ahmad et al. (2025) extract sparse rank-1 direction-level circuits via learnable masks on the singular values for the GP, IOI, and Greater-Than tasks.

We extend the analysis from individual directions to the 2D plane they span. The plane supports a continuous-angle injection $\alpha\sigma(\cos\theta u_i +$

$\sin \theta u_j$), which we characterise by phase stability across α together with a second-harmonic test that separates two-way antipodal axes from four-way categorical structure (§4.1).

Linear features in activation space. The linear representation hypothesis (Park et al., 2024), DAS (Geiger et al., 2024), geometry-of-truth probes (Marks and Tegmark, 2024), sparse autoencoders (Cunningham et al., 2023; Templeton et al., 2024), and function vectors (Todd et al., 2024) all search activation space for low-dimensional causal features. We search the weight matrix instead, which buys head-locality and removes probe training. The trade is that we cannot find features the head does not natively write through W_{OV} . DAS in particular is the closest neighbor: both methods identify a 2D rotation underlying behavior; DAS *learns* the rotation under causal abstraction objectives, the compass *reads* it from the OV SVD. We do not run DAS as a baseline; doing so would discipline our claim that the SVD-derived plane is the right one.

Helices and circular features. Engels et al. (2024) show LMs encode certain numerical concepts as approximate helices in the residual, recovered by trained probes. Our temporal dial in Phi-3 L24H17 (§4.1) is a categorical analog read from weights without probe training; an explicit head-to-head—do Engels’ months/days helix and our L24H17 plane occupy the same subspace?—would settle whether we are finding a different object or a re-derivation of the same one.

Steering primitives. ROME (Meng et al., 2022), ActAdd (Turner et al., 2023), and Inference-Time Intervention (Li et al., 2023) are 1D additive steering directions. The compass is a rank-2 generalization with a continuous- θ family, but—per the random-plane null result—we do not yet have evidence the rank-2 dial supports qualitatively new affordances over scalar 1D steering. The storage-steering dissociation (§4.4) suggests one place where it might.

Bias-evaluation context. We evaluate downstream behaviour on WinoGender (Rudinger et al., 2018; Zhao et al., 2018), intra-sentence StereoSet (Nadeem et al., 2021), and CrowS-Pairs (Nangia et al., 2020), with INLP (Ravfogel et al., 2020) and SentenceDebias (Liang et al., 2020) as the natural direction-based debiasing baselines. Two methodological caveats shape our reporting: Gonen

and Goldberg (2019) show linear-direction debiasing can move surface metrics while leaving downstream signals in nearby clusters, and Blodgett et al. (2021) document annotation noise in StereoSet and CrowS-Pairs of order a few SS points. We therefore present compass rotation as a graded steering primitive with an off-switch (§5.3) rather than as a debiasing method, and read movement magnitudes in light of the benchmark noise floor.

3 Method

3.1 Compass plane

For attention head h at layer ℓ we form $W_{OV}^{(\ell,h)} = W_V^{(\ell,h)} W_O^{(\ell,h)}$ and compute its SVD $W_{OV} = U \Sigma V^\top$, indexed by $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$. Following the convention in §2 we use rows of $V^\top$ as write-out directions and write $u_i := V_{[i,:]}^\top$ through-out. A *compass plane* is an ordered pair $\mathcal{C}_{i,j} = \text{span}(u_i, u_j)$ with $i \neq j$.

3.2 Causal α -sweep

The compass test is a causal injection that varies an angle θ and a scale α . Algorithm 1 states the procedure; Equation (1) gives the injection.

$$v(\theta, \alpha) = \alpha \sigma_i \cos \theta u_i + \alpha \sigma_j \sin \theta u_j. \quad (1)$$

Algorithm 1 Causal α -sweep on a compass plane.

Require: model M , layer ℓ , head h , dim pair (i, j) , probe tokens (t_+, t_-) , prompt set $\mathcal{P}$, angles $\Theta = \{0, \frac{2\pi}{36}, \dots\}$, scales $\mathcal{A} = \{1, 3, 10\}$

- 1: $W_{OV} \leftarrow W_V^{(\ell,h)} W_O^{(\ell,h)}$; $U \Sigma V^\top \leftarrow \text{SVD}(W_{OV})$
- 2: $u_i, u_j \leftarrow V_{[i,:]}^\top, V_{[j,:]}^\top$; $\sigma_i, \sigma_j \leftarrow \Sigma_{ii}, \Sigma_{jj}$
- 3: **for** $\alpha \in \mathcal{A}, \theta \in \Theta, p \in \mathcal{P}$ **do**
- 4: Hook blocks. ℓ .hook_resid_pre: $r \leftarrow r + v(\theta, \alpha)$ on the last token only
- 5: Forward pass; record $\text{LD}(\theta, \alpha; p) = \text{logit}_{t_+} - \text{logit}_{t_-}$
- 6: **end for**
- 7: Average LD over $\mathcal{P}$
- 8: Fit $\text{LD}(\theta) \approx \mu + A \cos(\theta - \phi)$ via first DFT bin
- 9: **return** (μ, A, ϕ) for each α

We average each LD across three prompts per condition (neutral, +, - context), so each α -sweep produces a 36-point curve from which (μ, A, ϕ) are read.

The two pillars. A causal compass passes both: amplitude linearity ($A(\alpha) \propto \alpha$, $R^2 \geq 0.95$ through the origin) and phase stability ($|\phi(\alpha_{\max}) - \phi(\alpha_{\min})| \leq 10^\circ$). Phase stability is the discriminating test: a generic 2D direction in the residual

will show some amplitude under linear injection (the linearity pillar partly tracks the linear injection itself), but only a direction the head actually writes through produces a phase that is invariant to scale. We require both pillars together because the joint constraint is rare in random subspaces: random orthonormal pairs in our scans pass linearity routinely but rarely hold phase below 10° .

3.3 Falsification battery

We complement Equation (1) with five pre-specified tests (verdicts in parentheses; $\checkmark$ = signal preserved, $\times$ = signal killed as a falsification target): **decode-heads** ($\checkmark$): $W_U^\top u_i$ promotes readable categorical tokens at $\theta \in \{0, \pi/2, \pi, 3\pi/2\}$; **cyclicity** ($\checkmark$): top-10 decode at $\theta = 0$ matches $\theta = 2\pi$ and is Jaccard-disjoint from $\theta = \pi$; **permutation null** ($\times$): permuting $\pm$ -context labels across prompts must kill the signal — this rules out the artefactual reading that the sinusoidal response merely tracks the prompt set rather than the categorical contrast; **random-plane null** ($\times$): random orthonormal directions in $\mathbb{R}^{d_{\text{model}}}$ should remove the causal effect; **causal-patch** ($\checkmark$): rotating the residual’s *natural* in-plane projection by $\Delta \in \{\pm 90^\circ, \pm 180^\circ\}$ preserves magnitude and shifts logits in the predicted direction.

The random-plane null is the most discriminating of the five in principle; §4.3 reports its outcome across the four-model panel and uses the result to scope the central claim.

3.4 Models and target heads

We use TransformerLens (Nanda and Bloom, 2022) with HookedTransformer on bfloat16. Targets:

- **GPT-2 Small** (Radford et al., 2019): L9H7 (u_1, u_2) (decode-discovered) and L10H9 (u_0, u_3) (scan-discovered).
- **Phi-3 Mini** (Abdin et al., 2024): L24H10 (u_0, u_1) (storage-dominant); L28H1 (u_0, u_3) (steering-dominant); L24H17 (u_1, u_2) (temporal); L24H28 (u_3, u_7) (entity).
- **Gemma-2-2B** (Gemma Team, 2024): L21H4 (u_0, u_2).
- **Llama-3.2-3B** (Grattafiori and Llama Team, 2024): L26H14 (u_2, u_3) (scan-discovered).

The blind-scan amplitude threshold is $A_{\text{slope}} \geq 0.20$ for GPT-2, Phi-3, and Llama, and 0.08 for Gemma. Gemma’s residual is normalized more aggressively after pre-norm, so per-unit α writes a

smaller fraction of the residual; using the model-specific threshold recovers the same head as the unit-norm injection. The single-head story does not change at $A_{\text{slope}} = 0.20$ either; the lower threshold widens the scan, it does not flip the identity of the discovered head.

4 Results

We organize results around a single question: when does the (u_i, u_j) plane behave like a causal compass, and when does it not? We start with the cleanest positive case (the Phi-3 temporal dial, §4.1), then move to the four-model gender panel (§4.2), then to the negative results: random-plane null failure (§4.3) and the storage–steering split (§4.4). The blind causal scan and Phi-3’s two gender heads are in §4.6 and §4.5.

4.1 Phi-3 temporal dial: a four-way categorical compass

The Phi-3 Mini head L24H17 with (u_1, u_2) exhibits the cleanest geometric reading of any compass in the paper. The two poles decode to temporal units — *month, months, mois* on one side, *hour, hours, evening* on the other — and the plane carries further structure on the orthogonal axis: *year, years, days*.

Two-harmonic test. A four-token probe at $\theta \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$ with tokens (*month, upon, hour, year*) lets us fit $f(\theta) = \mu + A_1 \cos(\theta - \phi_1) + A_2 \cos(2\theta - \phi_2)$. On the orthogonal axis $h(\theta) = \ell_{90^\circ} - \ell_{270^\circ}$, the temporal dial reaches $A_2/A_1 = 0.32$ at $\alpha = 10$: a substantial second harmonic, indicating a genuine four-way categorical structure. The Phi-3 entity dial L24H28 (*item* vs. *operation*) under the same probe scores $A_2/A_1 \leq 0.06$, consistent with two superposed antipodal axes. The two geometries are indistinguishable from decoding alone; the harmonic test discriminates them directly.

This is the central evidence that pairs of singular directions encode geometry that single directions cannot. Single-axis decoding of u_1, u_2 (or u_3, u_7 for L24H28) yields similar temporal vocabulary in both cases. The four-way structure emerges only on the joint plane. Full curves and per- α harmonic fits are in Section D.

4.2 Four-model gender panel

Table 1 reports the gender-pronoun compass across our four models. Phase drift is small ($\leq 3^\circ$) for

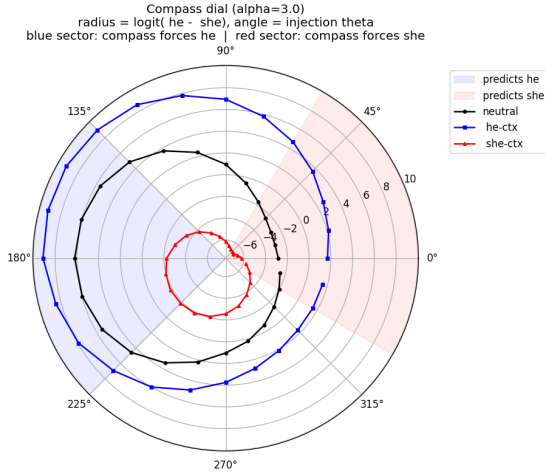


Figure 1: Polar plot for GPT-2 L9H7 at $\alpha = 3$. Radius: $\logit(he) - \logit(she)$. Angle: injection θ . Three concentric loops correspond to neutral (black), +context (blue), -context (red) prompts. The other three models are in Section E.

every head; baseline LD separations confirm the prompts are gender-discriminating before any intervention.

Decoding the basis. Every head decodes to a pronoun antipode. GPT-2 L9H7 splits *his/He/he* vs. *herself/her/She*; Phi-3 L28H1 splits *his/him/he/lui* (English plus French) vs. *her/she/hers/elle/ella* (English plus French and Spanish); Gemma L21H4 and Llama L26H14 produce similar antipodes with mixed non-English tokens. The cross-lingual decode is suggestive but we do not run within-language causal evaluation; it is a vocabulary overlap, not a multilingual finding (Section F).

4.3 The random-plane null fails for every primary head

The most discriminating test in our battery is the random-plane null: replace (u_i, u_j) with a random orthonormal pair drawn uniformly from $\mathbb{R}^{d_{\text{model}}}$, recompute the discriminative statistic $D = \overline{\text{LD}}_+ - \overline{\text{LD}}_-$ at $\alpha = 1$, and report $p = \Pr(D_{\text{rand}} \geq D_{\text{obs}})$ over 1000 trials. Results:

- GPT-2 L9H7: $D_{\text{obs}} = 25.49$, random-mean 26.58, $p = 0.380$
- Phi-3 L24H10: $D_{\text{obs}} = 22.59$, random-mean 23.10, $p = 0.352$
- Phi-3 L28H1: $D_{\text{obs}} = 40.77$, random-mean 20.47, $p = 0.142$

- Gemma L21H4: $D_{\text{obs}} = 51.89$, random-mean 26.59, $p = 0.115$
- Llama L26H14: $D_{\text{obs}} = 55.08$, random-mean 29.99, $p = 0.107$

None passes at $p < 0.05$. The Holm–Bonferroni step-down at family $\alpha = 0.05$ rejects no head: the smallest p (0.107) faces a threshold of $0.05/5 = 0.010$, and the rest face larger thresholds on weaker evidence (Table 2). For two heads (GPT-2 L9H7, Phi-3 L24H10) the observed signal is at the random-plane mean; for the remaining three the observed sits about $2\times$ the mean but inside a long tail.

Reading the result. The compass passes every other test in the battery (Table 3) — decoding, cyclicity, permutation null at $p < 0.001$, and the joint phase-stability + cyclic-decode criterion — and the random-plane mean signals cluster at 20–30 across all four models, not near zero. Taken together, these point to a specific geometry of the residual: gender is encoded over a distributed 2D subspace of the late-layer residual, and the OV-SVD plane is a weight-readable coordinate system inside that subspace. The plane the head writes through is well-defined and supports the causal interventions we run; the random-plane result tells us how widely gender content is spread across the residual, not that the head’s plane is uninformative.

A direct test of the distributed-subspace reading — decoding the random planes that exceed each head’s observed statistic and checking whether they too produce coherent gender antipodes — is a natural follow-up.

4.4 Storage versus steering

A second observation refines the picture. Different compasses respond very differently to two related tests. The additive α -sweep *pushes* the residual along the plane, regardless of whether the model already writes there. The in-plane causal-patch test *rotates the model’s natural in-plane projection* by $\pm 90^\circ$ or $\pm 180^\circ$, preserving magnitude. The two quantities dissociate (Table 4).

We use the labels *steering-dominant* and *storage-dominant* descriptively, not as a binary classifier; with $n = 5$ heads we treat this as a hypothesis to be tested at scale. The dissociation predicts downstream behavior: steering-dominant heads (Phi-3 L28H1, Llama L26H14) drive WinoGender stereo correlation efficiently (§5.1); storage-dominant

Table 1: Gender compass causal sweep. LD_+/LD_- : baseline $\text{logit}(t_+) - \text{logit}(t_-)$ on $\pm$ -context prompts. $A(\alpha)$: fitted sinusoidal amplitude. $\Delta\phi$: phase drift across $\alpha \in \{1, 3, 10\}$. Phase ϕ is in the plane’s own basis, so cross-model ϕ -comparisons are gauge-dependent (Llama L26H14 sits at $+161.6^\circ$ in our basis; the substantive cross-model claim is $\Delta\phi$, not ϕ).

Model	Head	(u_i, u_j)	(σ_i, σ_j)	LD_+	LD_-	$A(1)/A(3)/A(10)$	$\phi \pm \Delta\phi$
GPT-2 Small	L9H7	(1, 2)	(8.87, 8.46)	+5.31	-4.31	1.67/4.60/12.16	$-178.0^\circ \pm 2.9$
GPT-2 Small	L10H9	(0, 3)	(9.15, 6.80)	+3.73	-3.31	1.78/5.10/14.16	$+4.7^\circ \pm 0.3$
Phi-3 Mini	L24H10	(0, 1)	(9.42, 7.70)	+7.08	-6.97	0.67/2.01/5.22	$-154.9^\circ \pm 2.0$
Phi-3 Mini	L28H1	(0, 3)	(13.67, 7.61)	+5.06	-5.21	1.81/5.15/12.85	$+1.0^\circ \pm 1.3$
Gemma-2-2B	L21H4	(0, 2)	(2.35, 1.98)	+5.38	-5.21	0.18/0.59/2.19	$+0.2^\circ \pm 1.3$
Llama-3.2-3B	L26H14	(2, 3)	(1.14, 0.97)	+5.11	-4.58	0.28/0.88/3.01	$+161.6^\circ \pm 2.4$

Table 2: Holm–Bonferroni step-down on the five random-plane null p -values at family $\alpha = 0.05$. No head clears its threshold.

Head	p	Holm threshold	Reject?
Llama L26H14	0.107	0.010	no
Gemma L21H4	0.115	0.013	no
Phi-3 L28H1	0.142	0.017	no
Phi-3 L24H10	0.352	0.025	no
GPT-2 L9H7	0.380	0.050	no

Table 3: Falsification battery. cyc: cyclicity verdict. perm- p , rand- p : p -values from the permutation null and the random-plane null. Decode: top-10 vocabulary at $\theta = 0$ vs. π are categorically antipodal.

Head	cyc	perm- p	rand- p	decode
GPT-2 L9H7	✓	0.000	0.380	✓
Phi-3 L24H10	✓	0.000	0.352	✓
Phi-3 L28H1	✓	0.000	0.142	✓
Gemma L21H4	✓	0.000	0.115	✓
Llama L26H14	✓	0.000	0.107	✓

heads (Phi-3 L24H10, Gemma L21H4) sit closer to where the model naturally writes the category. The dissociation also explains why the random-plane null fails differently across heads: storage-dominant heads share their plane with much of the surrounding gender subspace, so a random plane has more chance of hitting it.

4.5 Phi-3 hosts two gender compasses on non-overlapping subspaces

Phi-3 L24H10 and L28H1 are both gender compasses, but they are not the same dial seen at two layers. Principal angles between the two 2D planes are $\{86.4^\circ, 89.0^\circ\}$ (essentially orthogonal). Their decode tables differ: L24H10 SV(0,1) decodes singular masculine vs. plural pronouns (*his/their*), while L28H1 SV(0,3) decodes singular masculine vs. singular feminine (*his/her*). The same model carries the gender axis at two distinct sites that sup-

Table 4: Storage vs. steering across five primary heads. $A_{\text{slope}} = A(10)/10$: α -sweep slope. Large $\Rightarrow$ plane is causally pushable (steering-friendly). $|\Delta_{\text{patch}}|$: maximum logit shift under in-plane rotation. Large $\Rightarrow$ model already writes substantially along the plane (storage-friendly). The two columns dissociate, with several heads sitting well off the diagonal.

Head	A_{slope}	$ \Delta_{\text{patch}} $	profile
GPT-2 L9H7	1.22	0.065	balanced
Phi-3 L24H10	0.52	0.190	storage-dominant
Phi-3 L28H1	1.29	0.024	steering-dominant
Gemma L21H4	0.22	0.316	storage-dominant
Llama L26H14	0.30	0.017	steering-dominant

Table 5: Blind causal-scan pass rates. Aggregate compass pass rate is at or just above the random-plane null on three of four models; the discriminating statistic is head-level concentration, not aggregate rate (see prose).

Model (planes / null)	Compass	Random null
GPT-2 (864 / 432)	1.9%	2.1%
Phi-3 (1344 / 672)	1.0%	1.3%
Gemma (240 / 120)	$\sim 2.0\%$	$\sim 1.6\%$
Llama (4032 / 2016)	0.22%	0.15%

port different rank-2 readings of it; this matches the analogous two-head pattern in GPT-2 (L9H7 storage-balanced, L10H9 storage-dominant).

4.6 Blind causal scan

To rule out the concern that compasses are surfaced by decoding-driven selection bias, we scan all $(\text{layer}, \text{head}) \times \binom{4}{2}$ planes formed from the top-4 SVs of each head’s W_{OV} : 864 planes for GPT-2, 1344 for Phi-3, 240 for Gemma, 4032 for Llama. The scan applies the two-pillar test ($R^2 \geq 0.95$, $\Delta\phi \leq 10^\circ$, $A_{\text{slope}} \geq 0.20$ except 0.08 for Gemma) without reading any decode table.

Table 5 shows the aggregate rates. On GPT-2 and Phi-3 they are within noise of the random null; on Llama they barely separate. The aggregate test does

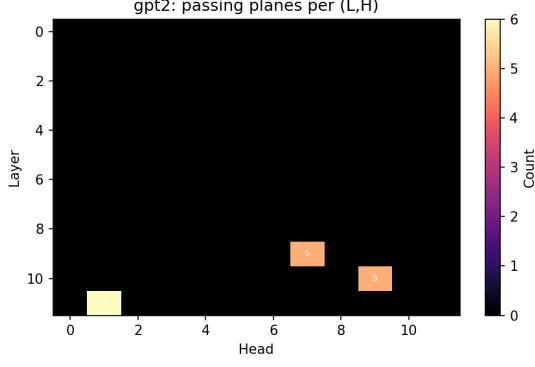


Figure 2: Blind causal-scan heatmap for GPT-2 (12×12 heads, 6 SV-pairs each, 864 planes). Three heads concentrate 15 of 16 passes; random-plane null planes distribute uniformly.

not distinguish the compass pipeline from random selection at $p > 0.05$ on three of four models.

Concentration is the discriminating statistic. What separates compass from random is per-head concentration. Of GPT-2’s 16 passing planes, 15 come from three heads (L9H7, L10H9, L11H1); on Llama, 7 of 9 passing planes come from two heads (L26H14 with 6/6 pairs, L20H13 with 3/6). Random-plane passes do not cluster this way. We acknowledge the limit of this argument: it is post-hoc on the choice of secondary statistic. Figure 2 makes the GPT-2 concentration visible.

The scan also surfaces heads the decode survey missed—GPT-2 L10H9, Phi-3 L28H1, Llama L26H14—validating it as complementary discovery.

5 Downstream behavior

We evaluate downstream behaviour on three benchmarks (WinoGender, single-head StereoSet, held-out routed CrowS-Pairs), framing compass rotation as a graded steering primitive with an off-switch. The reported movements are surface-metric shifts, in the sense of [Gonen and Goldberg \(2019\)](#); whether they correspond to the model encoding gender less or to a redirection of the same signal is a question for a follow-up probe-recovery experiment. Movement magnitudes should be read against the benchmark annotation-noise floor reported by [Blodgett et al. \(2021\)](#).

5.1 WinoGender

We sweep $\alpha \in \{0.25, 0.5, 1.0, 1.5, 2.0\}$ on Phi-3 L28H1 and report *stereo corr*, the Pearson

Table 6: WinoGender on Phi-3 Mini, L28H1 (u_0, u_3). ppl Δ : Wikitext-2 perplexity change.

Condition	raw gap	stereo corr	ppl Δ
baseline	1.151	+0.638	+0.000
ActAdd $\alpha=1.0$	3.506	+0.435	+0.139
compass $\alpha=0.25$	1.395	+0.640	+0.006
compass $\alpha=0.50$	1.743	+0.619	+0.006
compass $\alpha=1.00$	2.713	+0.468	+0.006
compass $\alpha=1.50$	3.937	−0.017	+0.018
compass $\alpha=2.00$	5.288	−0.303	+0.036

Table 7: StereoSet on Phi-3 L28H1. ActAdd outperforms compass on gender ICAT at the same head. We attribute this to L28H1 being steering-dominant (Table 4): natural StereoSet inputs project weakly onto its plane, so a single-head injection moves write weakly. Multi-head routed intervention closes part of this gap on CrowS-Pairs (§5.3).

Cond.	Domain	LMS	SS	ICAT
baseline	gender	92.75	73.33	49.46
compass $\alpha=1.0$	gender	92.94	73.73	48.84
compass $\alpha=1.5$	gender	92.75	73.73	48.74
ActAdd $\alpha=1.0$	gender	92.75	69.80	56.01
baseline	race	92.36	61.64	70.85
compass $\alpha=1.0$	race	92.41	62.06	70.13
ActAdd $\alpha=1.0$	race	92.57	61.43	71.40
baseline	profession	92.04	67.65	59.54
compass $\alpha=1.0$	profession	91.60	68.64	57.45
ActAdd $\alpha=1.0$	profession	91.73	64.44	65.23

correlation between the model’s per-occupation $\text{logit}(he) - \text{logit}(she)$ and the BLS female-occupancy rate; lower is less stereotypical, 0 is uncorrelated, negative is the opposite stereotype.

The compass intervention drives stereo corr through zero into reverse-stereotypical territory at low perplexity cost (+0.018 at $\alpha = 1.5$, vs. +0.139 for ActAdd at a comparable raw-gap shift). The reverse-stereotypical regime ($\alpha \geq 1.5$) is bias of opposite sign, not less bias; the table shows the full sweep so the operating point is the user’s choice, and we make no claim that any single α is the right one. A principled rule for picking α without a user-specified target is open.

5.2 StereoSet at single heads

Table 7 reports StereoSet LMS / SS / ICAT on Phi-3 L28H1. We report the result against ourselves: the single-head compass does not improve ICAT, and ActAdd outperforms compass on this benchmark.

5.3 CrowS-Pairs: held-out routed intervention

For multi-domain CrowS-Pairs we use a routed ensemble: the top-4 heads passing the multi-domain scan in each bias domain, each injected at SNR-calibrated $\alpha_d = \tau / \sqrt{\text{SNR}_h}$ with target τ chosen per model (0.20 GPT-2, 0.08 Phi-3 and Gemma, 0.10 Llama).

The held-out procedure. We split CrowS-Pairs 50/50 (seed 0) by domain. On the dev half we sweep a per-cell multiplier $M \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$ and pick M^* minimizing $|\text{SS} - 50|$ on the dev half. We then report SS on the test half. Including $M = 0$ in the sweep gives the procedure an off-switch when the baseline is already neutral.

Three patterns. (1) *Where the baseline is neutral, the procedure picks $M^* = 0$.* GPT-2 race ($M^* = 0$, test SS = 50.00); Gemma gender (test $|\Delta_{50}| = 0.38$); Llama race (2.71); Llama gender (3.44). Direction-based projector baselines do not have this property. Across the four-model panel $M^* = 0$ is selected on 10 of 16 cells, and on 5 of those 10 the test-half $|\Delta_{50}|$ is small (< 5); on the other 5 the dev sweep found no helpful multiplier and the test half remains biased (Phi-3 religion test 14.15, Gemma profession 18.60, Llama religion 31.13). $M^*=0$ is therefore an off-switch in the former cases and a no-help signal in the latter.

(2) *Where the baseline is biased, the compass moves it modestly, and the natural between-split comparison is missing.* On Phi-3 gender, the dev half $|\Delta_{50}|$ improves from 8.78 at $M = 0$ to 0.38 at $M^* = 0.50$; on the held-out test half the same $M^* = 0.50$ lands at $|\Delta_{50}| = 5.73$. The apples-to-apples comparison—test-half SS at $M = 0$ vs. test-half SS at $M^* = 0.50$ —requires running the $M=0$ condition on the test half for cells where $M^* \neq 0$, which our experimental setup did not do. The strongest defensible reading of the gender row is therefore: the dev-selected M^* generalizes onto the held-out half at a smaller-but-non-zero distance from neutral. On GPT-2 gender, $M^* = 0$ is selected because dev SS at $M = 0$ is already at 51.91 (within 2 points of neutral). Even the dev-side movement is well inside the Blodgett et al. annotation-noise band; we report it as suggestive rather than as evidence of debiasing.

(3) *Profession and religion overshoot the dev target on the held-out side.* Phi-3 profession test

SS = 60.47 vs. baseline 57.56; Llama religion test SS = 81.13. These cells reveal genuine train/test instability in CrowS-Pairs at small n ; with religion $n = 53$ the per-cell 95% CI is roughly ± 13 points, and M^* chosen on a 26-row dev half is unreliable. We report these cells to show the held-out procedure is not a guaranteed win.

What the held-out evaluation supports. The procedure — per-domain head routing, SNR-calibrated α , dev-selected multiplier with a 0-option — is a methodological improvement over fixed-projector baselines on three properties: (i) it admits an off-switch when the baseline is already neutral; (ii) it generalises across the dev/test split for the gender cell on Phi-3 and the neutral cells on GPT-2 race, Llama gender, and Gemma gender; (iii) it composes with per-domain calibration so race / profession / religion can be steered independently. Direction-based projector baselines on the same metric are in Section H.

6 Discussion

What the rank-2 reading buys. Reading attention-head weights as 2D planes rather than as collections of independent singular directions exposes structure the single-direction view cannot. The Phi-3 temporal dial’s four-way harmonic ($A_2/A_1 = 0.32$ at $\alpha = 10$ vs. ≤ 0.06 on the rank-2 two-way entity dial) is invisible to single-axis decoding. The storage-steering dissociation (Table 4) is recovered at the head level by reading the pair $(A_{\text{slope}}, |\Delta_{\text{patch}}|)$ jointly. The held-out routed CrowS-Pairs procedure with an $M = 0$ off-switch supports graded categorical steering at low language-quality cost.

Adjacent comparisons. Three follow-ups would discipline the positive findings: DAS (Geiger et al., 2024) on the same head (does a learned 2D rotation recover (u_i, u_j) ?); an Engels-style helix probe (Engels et al., 2024) on Phi-3 L24H17 (does the temporal dial occupy the same subspace as a probe-derived helix?); and a sparse-autoencoder decomposition (Cunningham et al., 2023; Templeton et al., 2024) of the same head (do two SAE features align with the compass plane?). All three are natural extensions of the geometric reading we report.

Limitations

The OV-SVD compass passes the joint phase-stability and cyclicity tests, but the random-plane

Table 8: Held-out routed CrowS-Pairs, dev-selected multiplier M^* scored on test half. $SS = 50$ is unbiased; lower $|\Delta_{50}| = |SS - 50|$ is closer to neutral. $M^* = 0$ entries: dev preferred no intervention. With $n = 131$ on gender, the per-cell 95% CI on SS is roughly ± 8.5 points; sub-point movements should not be over-read.

Domain	Phi-3 Mini				GPT-2 Small			
	M^*	SS_{test}	$ \Delta_{50} $	n	M^*	SS_{test}	$ \Delta_{50} $	n
gender	0.50	44.27	5.73	131	0.00	45.04	4.96	131
race	0.00	55.04	5.04	258	0.00	50.00	0.00	258
profession	1.25	60.47	10.47	86	1.50	59.30	9.30	86
religion	0.00	64.15	14.15	53	0.25	60.38	10.38	53

Domain	Gemma-2-2B				Llama-3.2-3B			
	M^*	SS_{test}	$ \Delta_{50} $	n	M^*	SS_{test}	$ \Delta_{50} $	n
gender	0.00	49.62	0.38	131	0.00	53.44	3.44	131
race	0.00	58.91	8.91	258	0.00	52.71	2.71	258
profession	0.00	68.60	18.60	86	1.50	54.65	4.65	86
religion	1.25	62.26	12.26	53	0.00	81.13	31.13	53

null returns $p > 0.10$ on every primary head (Holm-corrected; Table 2); we read the plane as a weight-readable 2D slice of a distributed gender subspace rather than a unique axis, and a direct test — decoding the random planes that exceed each head’s observed statistic — remains future work. Our gender evaluation uses binary English pronoun contrasts; non-binary expression and language-specific pronoun systems are out of scope, and the cross-lingual decode tokens in Section F are vocabulary overlap, not a within-language causal evaluation.

Ethical Considerations

The compass primitive is dual-use: the same rank-2 rotation that moves WinoGender stereo correlation toward zero, applied at the opposite pole, moves it the other way. The intervention is also asymmetric (Table 4): on GPT-2, masculine-coded directions steer more strongly than feminine-coded ones, so a practitioner applying the same α symmetrically will overshoot in one direction and undershoot in the other.

Restricted scope. Downstream evaluation uses public benchmarks (Gender Pronoun, WinoGender, intra-sentence StereoSet, CrowS-Pairs). These do not generalise to deployment-style settings (long-form generation, RLHF reward modelling, agentic systems); surface SS movement is a weak indicator of deployed model behaviour. Practitioners using the compass for fairness purposes should run the Gonen and Goldberg (2019) probe-recovery diagnostic before reporting any debiasing outcome.

Binary framing. The compass “gender” axis we measure is operationalised through binary English pronouns. Deployment to systems serving non-

binary users, or to non-English language settings, requires within-population evaluation that we do not provide.

Release. We release scan logs, calibration JSONs, per-head battery results, and pinned commit hashes sufficient to reproduce every number in Tables 1 to 8. We do not release pre-computed steering tensors. The (model, layer, head, SV-pair) tuples in Table 1 are sufficient to reconstruct the intervention from public weights; the methodology is in the paper rather than gated on an artefact.

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box held only one”, “She picked up a single”.
 – (operation-biasing): “The surgeon completed a complex”, “The factory ran one more”, “The team executed a risky”.

B Battery Implementation

Random-plane null. For each primary head, we draw 1000 random orthonormal pairs (u'_i, u'_j) from $\mathbb{R}^{d_{\text{model}}}$ via QR decomposition of Gaussian matrices, recompute the discriminative statistic $D = \overline{\text{LD}_+} - \overline{\text{LD}_-}$ at $\alpha = 1$, and report the empirical $p = \Pr(D_{\text{rand}} \geq D_{\text{obs}})$.

Permutation null. For each head we permute the $\{+, -\}$ context label across prompts 2000 times and compute the same D . The permutation null tests whether the signal could come from random label assignment; it returns $p < 0.001$ for every primary head we test.

Cyclicity. We project $u_i + iu_j$ onto the unit circle in $\mathbb{C}$ and read off the top-10 tokens at $\theta = 0, \pi/2, \pi, 3\pi/2, 2\pi$ via $W_U^\top$. The verdict *CIRCLE: closed and with distinct antipodes* requires Jaccard identity at $\theta = 0$ and $\theta = 2\pi$, and Jaccard zero between $\theta = 0$ and $\theta = \pi$. All five primary heads satisfy this.

Causal-patch. For a probe prompt, we cache the residual at the head’s layer, project onto the compass plane, rotate the in-plane projection by $\Delta \in \{\pm 90^\circ, \pm 180^\circ\}$ preserving the magnitude, and replace the residual with the rotated version on the last token only. We report the maximum absolute logit shift across rotations.

C Two-pillar thresholds

The blind scan applies three thresholds: $R^2 \geq 0.95$ on the linearity fit, $\Delta\phi \leq 10^\circ$ on the phase across $\alpha \in \{3, 10\}$, and $A_{\text{slope}} = A(10)/10 \geq 0.20$. For Gemma-2-2B, we lower A_{slope} to 0.08 because the post-pre-norm residual at the layer of interest is normalized to a larger magnitude than in the other three models, so the per-unit α writes a smaller fraction of the residual; the natural $\alpha\sigma$ injection (rather than unit-norm injection) recovers the same compass at the higher slope. The threshold change does not add or remove heads from the four-model story.

D Non-pronoun compasses in Phi-3

Table 9 reports the α -sweep numbers for the Phi-3 temporal and entity compasses.

Table 9: Non-pronoun compasses in Phi-3 Mini.

Axis	Head	(u_i, u_j)	$A(1)/A(3)/A(10)$	$\Delta\phi$
Temporal	L24H17	(1, 2)	0.69/2.10/7.60	0.9°
Entity	L24H28	(3, 7)	0.29/0.88/3.06	0.0°

A two-harmonic fit $f(\theta) = \mu + A_1 \cos(\theta - \phi_1) + A_2 \cos(2\theta - \phi_2)$ on a four-token probe at $0^\circ/90^\circ/180^\circ/270^\circ$ distinguishes two-way antipodal dials from genuine four-way categorical structure. Phi-3 L24H17 (temporal) scores $A_2/A_1 = 0.32$ at $\alpha = 10$ on the orthogonal axis, indicating four-way structure (month / upon / hour / year); Phi-3 L24H28 (entity) scores ≤ 0.06 , consistent with two superposed antipodal axes.

E Polar plots for the remaining models

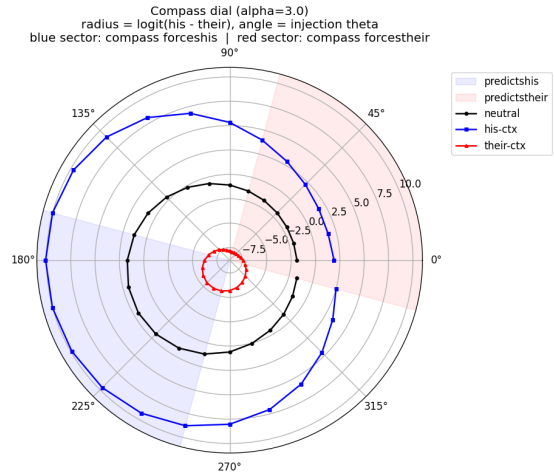


Figure 3: Phi-3 Mini, L24H10 gender compass, $\alpha = 3$.

F Decode tables (top-10 tokens per pole)

For each primary head, top-10 vocabulary tokens at the two compass poles via $W_U^\top u_i$:

GPT-2 L9H7 (u_1, u_2). Pole-0: *herself, her, She, she, Her, she, hers, She, hers, her*. Pole- π : *His, his, his, His, He, He, he, himself, his, He*.

Phi-3 L28H1 (u_0, u_3). Pole-0: *his, his, himself, lui (French), him, him, he, His, ego (Russian transliterated), ánico*. Pole- π : *her, she, hers, herself, she, elle (French), She, ella (Spanish), ela (Portuguese), her*.

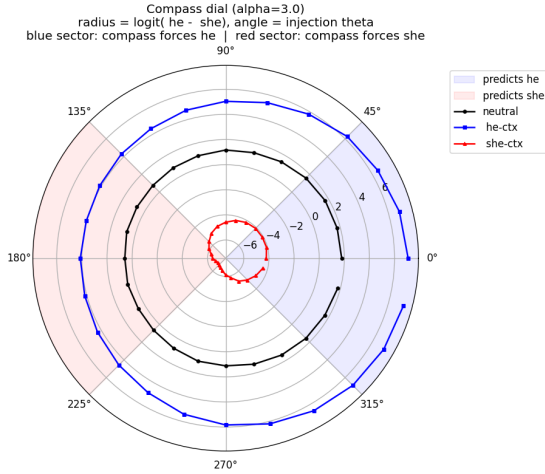


Figure 4: Gemma-2-2B, L21H4 gender compass, $\alpha = 3$.

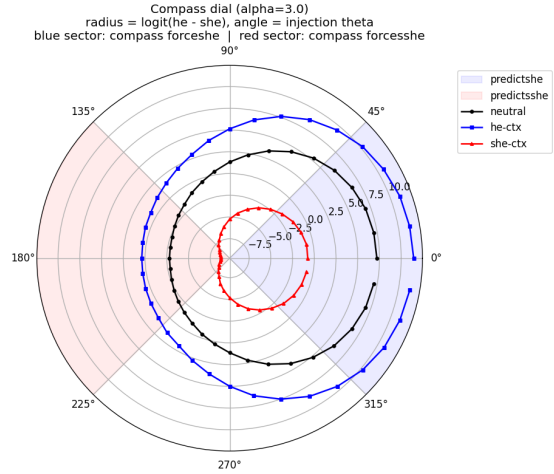


Figure 5: Phi-3 Mini, L28H1 gender compass, $\alpha = 3$.

Phi-3 L24H10 (u_0, u_1). Pole-0: *their, they, Their, loro, they, They, They, leur, leurs, them*. Pole- π : *Its, its, him, itself, suoi, its, ihm, jego, lui, ihm*. This plane decodes plural pronouns vs. mixed singular masculine forms with possessives in five languages; the contrast is more plural-vs-singular than masculine-vs-feminine, but the α -sweep on a he/she probe still passes the two-pillar test.

Gemma L21H4 (u_0, u_2). Pole-0: *him, kare, he, his, ihn, lui, his, him, his, He* (Japanese *kare* = “he”; German *ihn*; French *lui*). Pole- π : *she, her, she, She, herself, hers, She, her, she, geunyeo* (Korean *geunyeo* = “she”).

Llama L26H14 (u_2, u_3). Pole-0: *she, her, herself, hers, she, she, she, She, her, she*. Pole- π : *his, tā, his, himself, he, his, He, his, he, he* (Mandarin *tā* written transliterated; the underlying token in the unembedding vocabulary is the orthographic third-person pronoun).

G Per-axis decode for primary heads

For each primary compass head we list the top-8 vocabulary tokens at the positive and negative poles of its first few singular axes, decoded via $W_U^\top u_k$ with per-row centering of W_U . Reading axes individually gives an incomplete picture: e.g. Phi-3 L24H10 SV0 decodes to plural-vs-mixed-singular and SV2 to feminine-vs-masculine, but only the joint plane gives the gender axis the α -sweep finds. Non-Latin tokens are transliterated (Japanese *kare wa*, Ko-

rean *geunyeo*, Mandarin *ta*, Russian *shla/ono*, Portuguese *ela*, French *lui/elle/mois*, Italian *loro/egli*, German *ihn/ihm*, Indonesian *mereka*, Polish *jego*).

GPT-2 Small. L9H7. SV₀ ($\sigma=9.23$): *their, Their, they, themselves* vs. *his, He, him, His*. SV₁ (8.87): *herself, her, She, she, Her* vs. *His, his, He, he, himself*. SV₂ (8.46): *Its, its, itself, it, It* vs. *He, His, himself, his, herself*. SV₃ (7.69): non-categorical (token-fragment noise).

L10H9. SV₀ (9.15): *His, his, He, he, himself* vs. *her, she, She, herself, Her, hers*. SV₁ (7.87): *her, she, herself, She, haar* vs. *their, Their, they, They, THEIR*. SV₂ (7.27): *She, herself, her, she, hers, Her* vs. *Its, ITS, itself, it, It*.

Phi-3 Mini. L28H1. SV₀ (13.67): *his, himself, lui, him, he, His* vs. *her, she, hers, herself, elle, ella* — clean masculine/feminine. SV₁ (10.04): *his, him, himself, lui, ihm* vs. *Their, loro, leurs, themselves, Its*. SV₂ (9.27): *our, ourselves, notre, nous, ours* vs. *my, mys, My, myself*. SV₃ (7.61): *your, Your, yourself, yours* vs. *my, mys, My, our* — four orthogonal pronoun axes the same head writes through.

L24H10. SV₀ (9.42): *their, they, Their, loro, leur* vs. *Its, its, him, itself, suoi, jego* — plural-vs-singular. SV₁ (7.70): *its, Its, itself, Their, their* vs. *his, him, himself, ihn, ta* (Mandarin). SV₂ (6.94): *she, elle, shla, ella, her* vs. *him, his, lui, himself, ihm, ihn* — the gender axis. SV₃ (6.60): non-categorical vs. *she, hers, her, herself, ella, elle*.

L24H17 (temporal). SV₁: *month, months, mois*

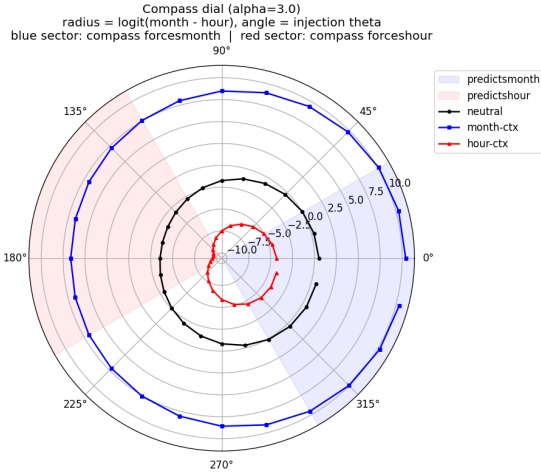


Figure 6: Phi-3 Mini, L24H17 temporal compass, $\alpha = 3$ (month vs. hour).

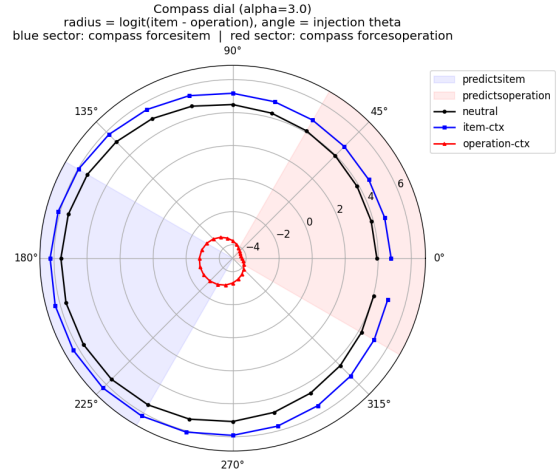


Figure 7: Phi-3 Mini, L24H28 entity compass, $\alpha = 3$ (item vs. operation).

vs. *hour, hours, evening, afternoon*. SV_2 : *year, years, days* (orthogonal axis).

L24H28 (entity). SV_3 : *items, atoms, objects, particles* vs. *operations, experiments, programs*. SV_7 : *teams, committees, authorities* (independent axis).

Gemma-2-2B L21H4. SV_0 (2.35): *him, kare wa, he, his, ihn, kare ga* vs. *she, her, herself, geunyeo*. SV_1 (2.07): *him, he, his, onun, her, ihm* vs. *they, mereka*. SV_2 (1.98): *him, he, his, egli, kare wa* vs. *its, ono, ta* (Mandarin) — *animate* vs. *inanimate*. SV_3 (1.52): *non-categorical* vs. *she, him, his, her, he*.

Llama-3.2-3B L26H14. SV_2 (1.14): *she, her, herself, hers* vs. *his, ta* (Mandarin), *himself, he*. SV_3 (0.97): *he, His, his, him, himself* vs. *her, herself, hers, ela*.

H Direction-based projector baselines on CrowS-Pairs

Table 10 reports INLP (Ravfogel et al., 2020) and SentenceDebias (Liang et al., 2020) alongside the routed compass at the $M^* = 1$ ($\alpha = \alpha_d$) operating point, on gender and race domains. INLP and SentDebias are trained on StereoSet intra-sentence labels at one layer per model; both projectors are applied as residual hooks. Neither has an off-switch; both modify already-neutral cells.

The compass column is the in-sample $M = 1$ result, before the held-out multiplier sweep of Ta-

Table 10: Projector baselines on CrowS-Pairs SS.

Model	Domain	baseline	INLP	SentDeb.	compass
GPT-2	gender	48.47	61.07	43.89	33.59
GPT-2	race	50.00	40.70	49.61	53.29
Phi-3	gender	57.63	63.74	53.05	45.80
Phi-3	race	52.52	50.39	50.97	34.30
Gemma	gender	51.15	58.78	54.20	32.44
Gemma	race	56.01	56.40	53.88	31.98
Llama	gender	54.20	57.25	51.53	55.34
Llama	race	50.97	46.90	45.16	45.35

ble 8. The held-out evaluation is the more honest read.

I Per-model α -sweep curves and linearity

The per-model curves below show $LD(\theta)$ for all three α values, paired with the $A(\alpha)$ linearity plot.

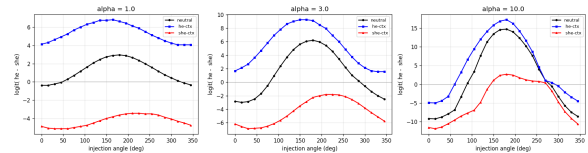


Figure 8: GPT-2 Small, L9H7 gender compass: $LD(\theta)$ across $\alpha \in \{1, 3, 10\}$.

J Llama-3.2-3B compass polar

K Blind-scan heatmaps and routing

The blind causal scan produces a (layer, head) heatmap of passing planes per model. Figure 2 (in the main paper) shows GPT-2; Figures 15 to 17

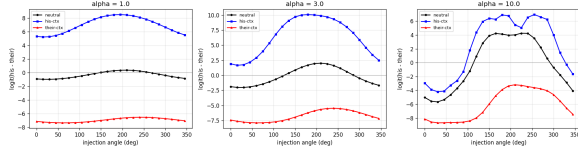


Figure 9: Phi-3 Mini, L24H10 gender compass: $LD(\theta)$ across $\alpha \in \{1, 3, 10\}$.

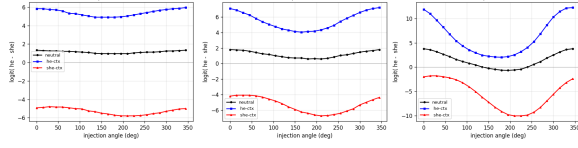


Figure 10: Gemma-2-2B, L21H4 gender compass: $LD(\theta)$ across $\alpha \in \{1, 3, 10\}$.

report the other three models. Figure 18 reports the multi-domain routing density used by the held-out CrowS-Pairs intervention.

L Per-domain α calibration

We calibrate $\alpha_d = \tau / \overline{\text{snr}}_h$ where $\text{snr}_h = \|v_h\| / \|r_{\text{pre}}\|$ at the head’s layer on a neutral prompt, and τ is a model-specific target SNR. The calibrated α_d values used for $M = 1$ in Table 8 are: GPT-2 ($\tau = 0.20$): 1.89/1.80/1.37/1.87 for gender/race/profession/religion. Phi-3 ($\tau = 0.08$): 5.10/5.15/3.80/5.34. Gemma ($\tau = 0.08$): 55.26/81.36/29.73/31.19 (Gemma’s smaller singular values require larger α to match SNR). Llama ($\tau = 0.10$): 4.06/3.56/3.45/4.66.

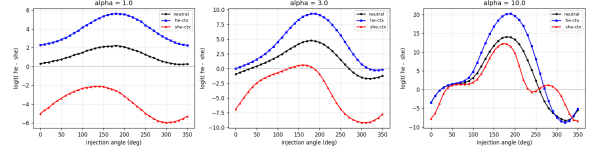


Figure 11: Llama-3.2-3B, L26H14 gender compass: $LD(\theta)$ across α .

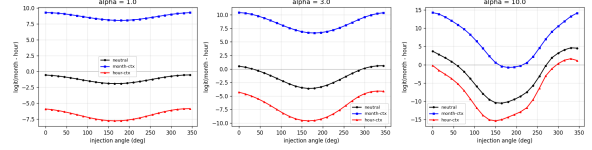


Figure 12: Phi-3 Mini, L24H17 temporal compass (month vs. hour): $LD(\theta)$ across α .

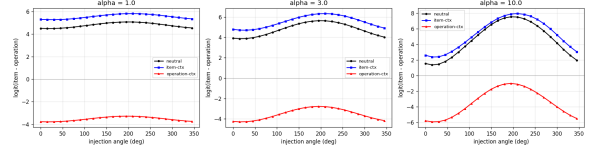


Figure 13: Phi-3 Mini, L24H28 entity compass (item vs. operation): $LD(\theta)$ across α .

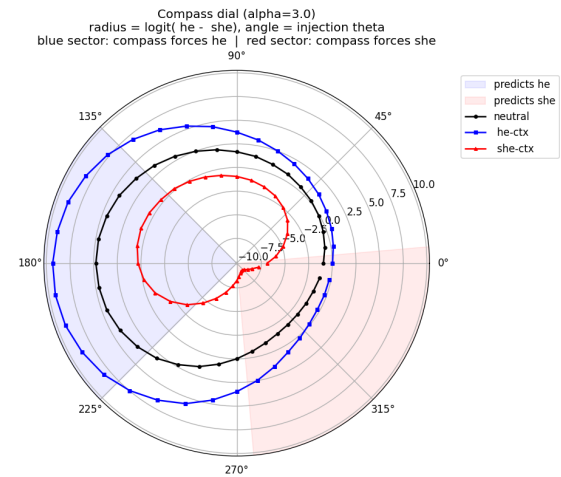


Figure 14: Llama-3.2-3B, L26H14 gender compass at $\alpha = 3$.

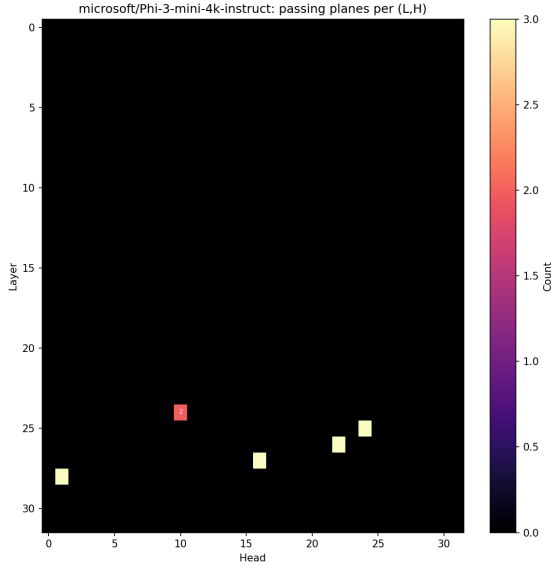


Figure 15: Phi-3 Mini blind-scan heatmap, gender domain.

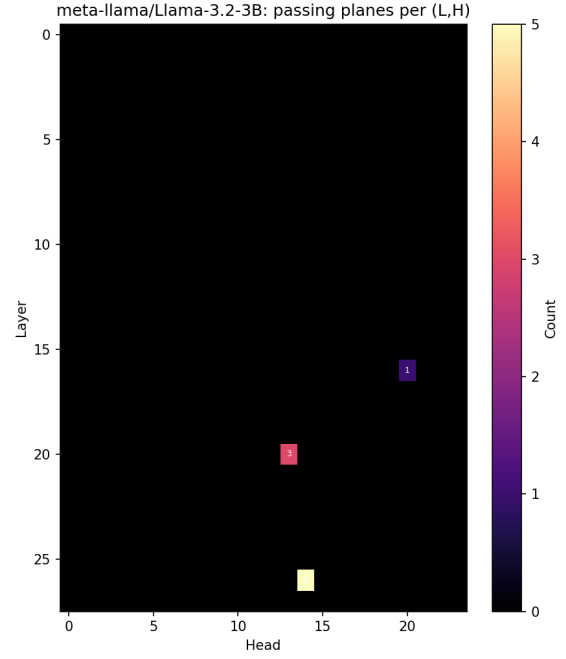


Figure 17: Llama-3.2-3B blind-scan heatmap, gender domain.

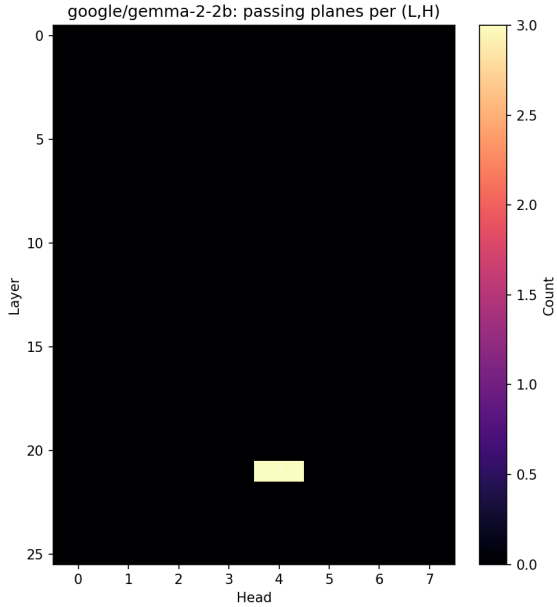


Figure 16: Gemma-2-2B blind-scan heatmap, gender domain ($A_{\text{slope}} \geq 0.08$).

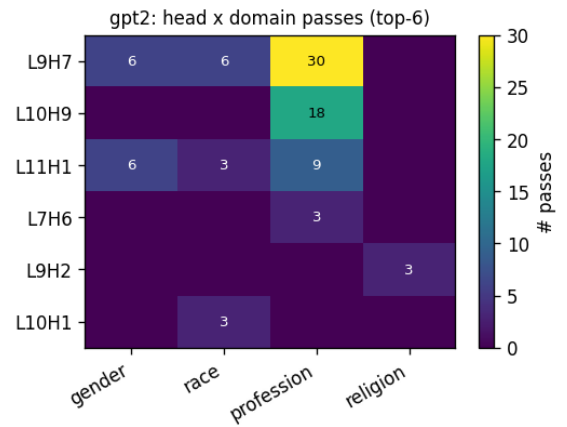


Figure 18: GPT-2: per-head pass count across the four CrowS-Pairs domains (gender, race, profession, religion). Heads passing on multiple domains are routed by domain specialisation in §5.3.

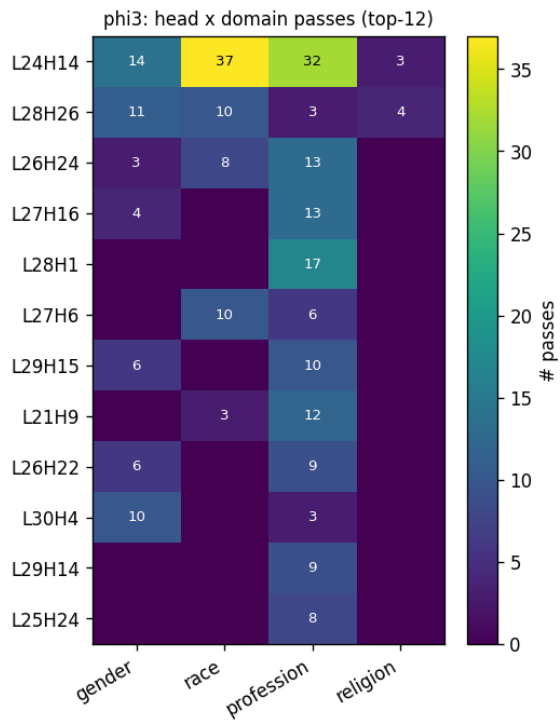


Figure 19: Phi-3 Mini: per-head pass count across CrowS-Pairs domains.

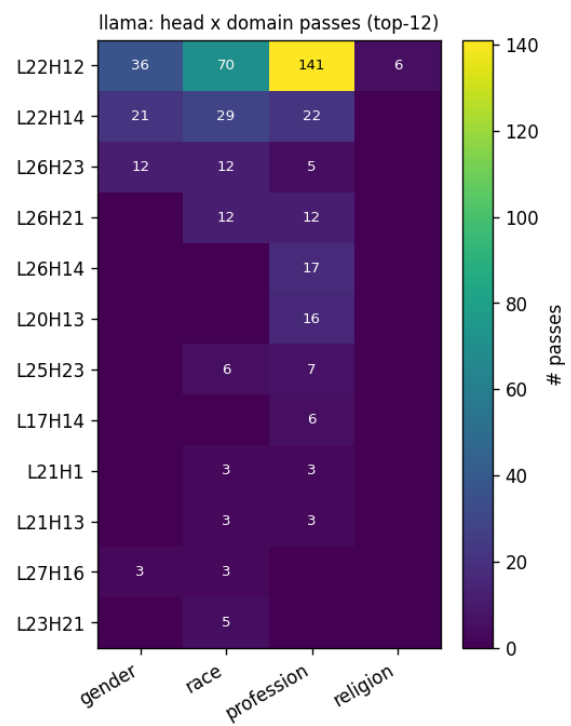


Figure 21: Llama-3.2-3B: per-head pass count across CrowS-Pairs domains.

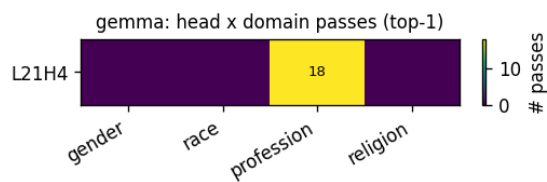


Figure 20: Gemma-2-2B: per-head pass count across CrowS-Pairs domains.